



Name of the module: cycShiftFD
Target delivery date:
Location in GIT repository: vspa_common\vspa-lib\cycShiftFD

Overview:

This module is used to perform the cyclic shift in frequency domain. The output is equivalent to cyclic shifting to the right with $N = \text{“shift_size”}$ samples in time domain.

Function API:

```
void cycShiftFD_asm( cfloat16_t *in_p,  
                    cfloat16_t *out_p,  
                    int          shift,  
                    int          length  
);
```

API description:

CycShiftFD operation is implemented via elementwise vector multiplication in time domain: $\mathbf{y} = \mathbf{x} \cdot \mathbf{m}$, where $\mathbf{x}$ is the input signal, $\mathbf{m}$ is an NCO-generated complex sequence, and $\mathbf{y}$ is the mixed signal.

Function inputs:

in_p	Pointer to input, the signal to be mixed. Vector-aligned. In complex 16-bit half-fixed.
shift_size	Samples to shift with
input_size	Number of input samples

Function outputs:

out_p	Pointer to output, the mixed output. Vector-aligned. In complex 16-bit half-fixed.
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DMEM requirements:

Variable	Minimum size (words)	Minimum alignment (words)	Allocated by
output	32	32 (vector-aligned on 16AU)	Caller
input	32	32	Caller
shift_size	1	1	Caller
input_size	1	1	Caller

Implementation:



The following is to explain how NCO uses “PhaseIn” and “FreqIn” to generate a complex sequence. CycShiftFD application configures NCO in “normal” mode. In 16AU, NCO configured in normal mode generates a 16-point complex exponential sequence **m** by:

$$\mathbf{m} = e^{\frac{j2\pi kf(i+16n+[0,1,2,...,15])}{2^{32}}},$$

where

- *k* is base frequency multiplier in uint16,
- *f* is base frequency in int32 with 1’s complement,
- *i* is initial phase in uint32,
- *n* is the batch index. Every read of S1 increases *n* by 1.

The implementation configures NCO as such:

Parameters of NCO	Value
<i>k</i>	1
<i>f</i>	shift_size * 2 ³² /input_size
<i>i</i>	0
<i>n</i>	starts with 0 when first entering the cycShiftFD function

Test plan:

Various sizes and shifts

References:

VSPA_16AU_ISA_v2.0_ISM, Section 4.20 “Numerically controlled oscillator (NCO) instructions”
mixer function in Rayleigh VSPA1-4AU library